

Introduction to Machine Learning

Group Assignment (2/3 members per group)

For the assignment, you are required to explore the application of Machine Learning techniques to the data problem of your choice. You may choose to study any data problem, giving special consideration to the unique properties of the problem domain, and testing one or more methods on it (**2 machine learning predictive models**). You may ensure that the dataset is *sufficiently challenging* so that there is room to demonstrate your creativity and knowledge in the subject to address these challenges.

Your analysis and implementation of models should be in considerable depth and detail, also it may *go further* than what has already been covered in this course.

Please note that you may need to pre-process your data to get it into an appropriate format. This assignment should involve a detailed exploration along with analysis for each of the experiment and outcome.

Learning Outcome

Demonstrate solutions obtained by applying appropriate machine learning models for various types of problems.

Assessment

The total assessment mark of this group assignment is 30%. Marking criteria is available along with this assignment.

Acquiring the datasets

This assignment must involve **ONE** dataset. The dataset should be of reasonable size. If your dataset is imbalanced, find ways to balance the dataset.

Some guidelines on choosing a dataset:

- Shall not be a perfectly clean dataset
- Shall contain categorical and numeric data
- Shall be reasonably large and (say) have more than 12 variables (a guideline)
- Not a commonly experimented dataset.

Deliverables

The assignment work must be in the form of a report. It should be submitted on **08.05.2026**. All submissions are **ONLINE via MOODLE**.

Please follow APU report format.

All reports must be typeset in 12pt font. **Line spacing at least 1.15 or 1.5** The text in your report must be in your own words. Quoted text must be set off by quotes (“ ”) and the source clearly attributed, even if the text is as small as a single phrase. If you use someone else's figures or tables, the appropriate attribution must appear in the caption as well as in the text where you discuss the figure/table. Please do not turn in large sections of code listings or massive tables of raw data. The report should be easy to read. Every figure or table in the report body should be discussed in the report body. If you would like to present additional experiments that are not evaluated in the body of your report, include them as an appendix.

The following also must be submitted together as a compressed file (.zip or .rar) with your report.

1. Original dataset
2. All source codes (.ipynb format)

The report may contain the following format.

Title, Abstract (weightage 2%)

The title of your report should indicate the learning problem it addresses.

Your report should start with a short 1-paragraph abstract that mentions the problem you addressed, two or three sentences on the background to the problem, its significance and how your work is positioned in the backdrop of the literature. It shall briefly mention any preprocessing performed, models implemented and a few sentences on the results and conclusion.

Abstract shall typically be half a page in length with a single paragraph.

Introduction, Aim & Objectives (weightage 3%)

The introduction should contain a description of your problem at a level that anyone could understand, any area-specific jargon should be explained. It can also give an overview of how you obtained your data, the URL etc. However, this additional information is likely to appear elsewhere and so should be just summarized in the introduction to avoid too much redundancy. If your problem is technical or difficult to describe precisely then you might give just an overview of it in the introduction and use a different section to describe the detailed questions you attempt to

answer. The introduction should provide an overview of what the problem is, why it is interesting/important (why did you choose it) and how you addressed the problem. This section shall contain Aim, objectives, and scope of your work.

This section shall typically be 1-2 pages in maximum.

Related works (weightage 5%)

The related work section should contain a survey of relevant previous works (the papers related to your selected problem/dataset) for your problem and the methods you used in your experiments. However, the best related work sections are not just lists of references, but evaluations. You should include references and results related to your choice of dataset. The experiments and results on the same dataset shall later be used for comparing with the results of your work. All references shall be finally summarized in a table form. Include one paragraph referring to this table and this summary shall assist/lead you in identifying the directions to your proposed work. This paragraph shall also clearly spell out what you did for the assignment in relation to what was done by others. Feel free to cite textbooks or articles etc. for descriptions of algorithms. The references should preferably be recent, (e.g.: later than 2016). A rough guideline for the number of references may be around 10-15.

The length of this section may be to a maximum of 4 pages.

Methods (2%)

This section should describe the details of your experiments to predict/classify according to the selected problem. It should start with a description of the data, including the data source, the number of examples, the features and labels.

Describe the learning techniques used and what software packages you used. You should also explain here *which machine learning methods you picked and why*. Also describe the metrics that you choose for the performance evaluation.

This section may be to a maximum of 2 pages

Dataset Preparation (weightage 3%)

Dataset

Full details of the dataset and complete exploratory data analysis of the dataset (EDA) shall be included.

To go through data selection, cleaning, formatting, and exploring. The goal of exploring is to identify the most important fields in predicting an outcome and determine which attributes may be useful. A complete exploratory data analysis (EDA) and pre-processing of the dataset is required.

Model implementation (weightage 8%)

Build the model/s you have chosen. It is suggested that you build at least 2 models. In applying a model, you shall include optimization (tuning) procedures. Comprehensively explain the output of the model and its internal parameters.

Model Validation (weightage 2%)

After building the models, you must evaluate and compare the results and interpret the significance, thus confirming that the outputs of the models are statistically acceptable and leading to select the most suitable model at last.

Analysis & Recommendations (weightage 3%)

Once a model is built and validated the results must be critically analysed. The experimental results section should describe what happened. Is it what was expected? What were the surprises/anomalies? In retrospect, why do you think the results come out the way they did? How do you compare your obtained results with the results reported in the related work that you have selected. Tables and graphs are appropriate ways to summarize the results of all the implemented models. You must include comparisons between the models implemented. Recommend actions based on your discussion of the results.

Conclusion (weightage 2%)

The conclusions section should include a short self-evaluation of your project of what went right and what went wrong, together with a summary of what was learned from the experiments and what you yourself learned and a recap of what you accomplished. If there are other things you would have liked to try but didn't get around to, you can include future work in the conclusions section or even do further work in its own section.

References

The references should contain relevant publications (articles, books, etc.) that you read in conjunction with your project. Most importantly it should include the references to the dataset as well as any kernels available online for this dataset. Part of the project is to identify the relevant literature and read in more detail about some aspects of machine learning. All references shall follow APA format.

Note: Bibliography and Acknowledgements are mandatory even though there is no specific mark allocated to them. You may also note that credit for the Bibliography section is covered by the related work section it supports.

Plagiarism

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